

S003 - Technical Integration Notes: Shopify Connector Bridge

Technical Notes | Engineering

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1. Scope of These Notes

These notes describe how GreenCart Organics (S003) is technically integrated with the platform, and record the behaviour of the integration during the two production incidents on this account in 2026. They are written for engineers who need to reason about where a fault lies when GreenCart reports a problem.

The central point is that GreenCart's integration has three layers, and most confusion on this account comes from attributing a fault to the wrong one.

Layer	Owned by	Typical failure signature
Shopify storefront and order objects	GreenCart	Missing or malformed order attributes
Connector bridge	Platform, shared operation	Translation errors, dropped metadata
Nexora platform services	Platform	Service errors, timeouts, event duplication

2. Order Intake Path

An order placed on the marketplace is delivered to the connector by webhook. The connector translates it into a Shopify order object, applies the delivery slot as a line-item property, and acknowledges. GreenCart's dark store systems read from Shopify.

Important characteristics:

- Delivery is **at-least-once**. The connector deduplicates on the platform order identifier before creating a Shopify order.

- Slot metadata travels as a line-item property, not as an order-level field.

This was a Shopify modelling constraint and is the reason the pilot found an amendment defect in August 2025.

- Acknowledgement is fast and processing is asynchronous, so a slow Shopify API does not cause webhook retries at the platform end.

3. Inventory Path

Stock is maintained per dark store. The connector publishes availability at slot granularity: a SKU can be available for tomorrow's morning slot and unavailable for tonight's, because stock is allocated against slot capacity rather than against a single pool.

This is unlike any other account on the platform. S001 and S002 publish a single availability figure per SKU per location. Engineers investigating a GreenCart availability question should not assume the S002 model applies.

4. Label Generation Path

Every dispatch requires a shipping label from the Shipment Label Service. The service is configured with a primary courier provider (SwiftLane) and a backup (RapidPost).

4.1 Behaviour Observed in SC-005, 18 June 2026

SwiftLane's API degraded during the morning slot cut-off. The expected behaviour was automatic failover to RapidPost. Failover did not occur.

The reason was that the stored RapidPost credentials had expired. The failover path attempted to authenticate at the moment failover was needed, the authentication failed, and the service fell back to retrying the primary provider rather than surfacing a credential error. 2,100 orders were left in label-pending state and 640 perishable orders breached their delivery slot.

Three engineering observations from that incident are worth recording.

1. The failover path was never exercised in production before it was needed, so an expired credential was indistinguishable from a working configuration.
2. Failing back to the degraded primary provider was worse than failing loudly.

The system preserved the appearance of function while producing no labels.

3. Credential expiry was observable in the credential store but was not monitored.

Remediation: credentials rotated, an automated expiry monitor added covering both providers, and a failover pre-flight check that validates backup credentials on a schedule rather than at the moment of need.

5. Refund Path

Refunds originate either from a platform-side action or from GreenCart's refund-on-report process for quality complaints. Both produce a refund event on the event bus, consumed by the refund webhook consumer.

5.1 Behaviour Observed in SC-007, 3 September 2026

The refund webhook consumer restarted during a deployment. The broker redelivered events that had been processed but not acknowledged. Because the consumer applied refunds without checking an idempotency key, 640 refunds were applied twice, an excess of INR 1,18,000.

GreenCart was disproportionately exposed to this defect for a structural reason: the refund-on-report category exception means this account generates a far higher ratio of refund events to orders than any other seller. The defect was platform-wide, but it surfaced here first and hardest.

Remediation: an idempotency store keyed on refund event identifier, with a retention window of 30 days; duplicate refunds reversed; consumer replay guarded; and a duplication alert configured on refund volume per account.

The requirement to tolerate at-least-once delivery was written into the S002 integration specification as ORD-3 and into the S001 integration design in 2023. The refund consumer predates both and was never revisited. This is a documentation and inventory problem as much as a coding defect.

6. Known Constraints

Constraint	Consequence
Slot metadata is a line-item property	Order amendments must re-apply the slot
Availability is slot-scoped	Single-figure availability queries are meaningless here
Refund-on-report exception	High refund event volume relative to order volume
Connector is a shared component	A connector change affects this seller immediately
No seller-side engineering capacity	Platform carries more integration responsibility here

7. Investigation Guidance

When GreenCart reports a problem, establish the layer first.

1. If the order exists on the platform but not in Shopify, the connector is the first place to look.
2. If the order exists in both but has lost its slot, check whether an amendment occurred.
3. If labels are not generating, check provider health and, separately, backup credential validity. These are different failures.
4. If refund amounts look wrong, check for duplicate application before assuming a calculation error.
5. If availability looks wrong, confirm which slot the query refers to.

8. Revision History

Version	Date	Change
1.0	2025-10-02	Initial notes after subscription slot model enabled
1.1	2026-06-26	Label generation path and SC-005 behaviour added
1.2	2026-09-22	Refund path and SC-007 behaviour added